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DEVELOPING A NOVEL ALGORITHM FOR BETTER INTERPOLATING LOW RESOLUTION MEDICAL IMAGES

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Paper Title	DEVELOPING A NOVEL ALGORITHM FOR BETTER INTERPOLATING LOW RESOLUTION MEDICAL IMAGES (/PublishedPaper/18Vol/Issue6/2026EUSRM6202661827-e3cd538b-fd24-4b52-8eed-756df944b62f27898.pdf)
Author Name	Mr. Shams Tabrez, Dr. Alka Thakur
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Abstract	Medical imaging modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Ultrasound (US), Positron Emission Tomography (PET), and X-ray often suffer from low spatial resolution because of hardware limitations, acquisition time constraints, patient movement, and radiation dose reduction requirements. Conventional interpolation techniques including Nearest Neighbor, Bilinear, and Bicubic interpolation produce blurred edges and fail to preserve fine anatomical structures that are essential for clinical diagnosis. Recent deep learning-based super-resolution methods achieve improved visual quality but generally require large annotated datasets, high computational resources, and extensive training time. This paper proposes a Novel Hybrid Edge-Aware Adaptive Medical Image Interpolation Algorithm (HEAMI) for enhancing low-resolution medical images. The proposed algorithm integrates adaptive edge detection, multi-scale feature extraction, hybrid convolutional learning, and
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